



1. Networking Fundamentals

1.1 Define network concepts

• Internet, intranet, extranet, client-server, peer-to-peer, transmission types (unicast, multicast, broadcast), network devices including IoT

2

Connecting to a private network address from a public network requires:

- A. Network address translation (NAT).
- B. Dynamic Host Configuration Protocol (DHCP).
- C. Network Access Protection (NAP).
- D. Dynamic domain name system (DDNS).

Correct: A

3

A Media Access Control (MAC) address identifies a/an:

- A. UPnP device.
- B. Local broadcast domain.
- C. Network interface card (NIC).
- D. Local area network (LAN).

Answer : C

4

Two companies want to share data by using the Internet.
Which type of network provides the solution?

- A. Ethernet
- B. Intranet
- C. Extranet
- D. Perimeter

Answer : C

5

Which technology can you use to extend an internal network across shared or public networks?

- A. VLAN
- B. Microsoft ASP-NET
- C. Microsoft .NET Framework
- D. VPN

Correct: D

6

One purpose of a perimeter network is to:

- A. Make resources available to the intranet.
- B. Link campus area networks (CANs).
- C. Link local area networks (LANs).
- D. Make resources available to the Internet

Correct: D

7

A private network that allows members of an organization to exchange data is an:

- A. Extranet
- B. Ethernet
- C. Intranet
- D. Internet

Correct Answer: C

8

Match each network type to its corresponding definition.

To answer, drag the appropriate network type from the column on the left to its definition on the right. Each network type may be used once, more than once, or not at all. Each correct match is worth one point.

Network Types	Answer Area
Extranet	a network that allows controlled access for specific business or educational purposes
Internet	a network that allows access only to users within an organization
Intranet	a system of interconnected networks

Correct Answer:

Extranet – a network that allows controlled access for specific business or educational purposes

Intranet – a network that allows access only to users within an organization

Internet – a system of interconnected networks

9

What technology takes data and breaks them into packets and sends them over a network, some times using different routes for each packet?

- a. circuit switching
- b. connection switching
- c. packet switching
- d. network switching

Correct Answer: C

1.2 Define cloud and virtualization concepts

10

What name does windows server 2008 R2 use for terminal service?

- a. MS telnet server
- b. MS primary services
- c. MS thin server
- d. remote desktop service

Correct Answer: D

11

What would you use to create VPN tunnels so that users can connect to your internal network while at home using windows server 2008 R2?

- a. Microsoft RAS
- b. Microsoft RRAS
- c. Microsoft RDC
- d. Microsoft VPN Server

Correct Answer: B

1.3 Describe remote access methods

12

VPNs are implemented to provide:

- A. A secure connection within a private network.
- B. A secure connection through public networks.
- C. Additional encryption by using IPSec.
- D. Additional security for selected computers.

Answer : B

13

A VPN is a/an:

- A. Encrypted connection across the Internet.
- B. Virtual network within your local area network (LAN).
- C. Communication tunnel between VLANs.
- D. Personal network for your use only.

Correct: A

14

This question requires that you evaluate the underlined text to determine if it is correct.

"A/anvirtual private network (VPN)" protects a network's perimeter by monitoring traffic as it enters and leaves.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. Extranet
- B. Firewall
- C. Intranet
- D. No change is needed

Correct Answer: B

15

Match the VPN connection type to the corresponding definition.

To answer, drag the appropriate VPN term from the column on the left to its definition on the right. Each term may be used once, more than once, or not at all. Each correct match is worth one point.

Terms	Answer Area
Point-to-Point Protocol	allows a remote user to connect to a private network from anywhere on the Internet Term
SSL VPN	securely connects two portions of a private network or two private networks Term
Layer 2 Tunneling Protocol	creates an unencrypted connection between two network devices Term
Site-to-Site VPN	

Correct Answer:

SSL VPN - allows a remote user to connect to a private network from anywhere on the Internet

Site-to-Site VPN – securely connects two portions of a private network or two private networks

Layer 2 Tunneling Protocol - creates an unencrypted connection between two network devices

16

What port does remote desktop services use?

- a. 443
- b. 501
- c. 389
- d. 3389

Correct Answer: A

17

You use the _____ to connect to a terminal server

- a. Remote Dekstop Connection (RDC)
- b. Remote Dekstop Protocol (RDP)
- c. Remote Session Call (RSC)
- d. Remote NetBios Connection (RNC)

Correct Answer: A

18

What allows a user to connect to a corporate network using the internet?

- a. VPN
- b. VPC
- c. PPTN
- d. IPSec

Correct Answer: A

19

What command do you use to connect to a remote computer so that you can execute commands?

- a. ftp
- b. nslookup
- c. telnet
- d. nbstatat

Correct Answer: C

2. Network Infrastructures

2.1 Define the characteristics of local area networks (LANs)

20

A network that separates an organization's private network from a public network is a/an:

- A. Firewall
- B. Extranet
- C. Perimeter
- D. Internet

Answer : C

21

In local area network (LAN) topologies, the primary media access methods are: (Choose two.)

- A. Contention
- B. Negotiation
- C. Kerberos
- D. Token passing

Correct: AD

22

In a physical star topology, the central device is referred to as a:

- A. Bridge
- B. Server
- C. segmenter
- D. Hub

Correct: D

23

The topology of a local area network (LAN) is defined by the:

- A. Number of devices to connect.
- B. Physical and logical characteristics.
- C. Distance between workstations.
- D. Type of cable being used.

Correct: B

24

A node within a local area network (LAN) must have a network interface device and a:

- A. Network account
- B. Table of all network nodes
- C. Host address
- D. Resource to share

Correct: C

25

Which type of port is used to support VLAN traffic between two switches?

- A. Virtual port
- B. WAN port
- C. Trunk port
- D. LAN port

Correct: C

26

One reason to incorporate VLANs in a network is to:

- A. Increase the number of available IP addresses.
- B. Increase the number of available Media Access Control (MAC) addresses.
- C. Reduce the number of broadcast domains.
- D. Reduce the number of nodes in a broadcast domain.

Correct Answer: D

27

Match the networking topologies to their corresponding characteristics.

To answer, drag the appropriate topology from the column on the left to its characteristic on the right. Each topology may be used once, more than once, or not at all. Each correct match is worth one point.

Topologies	Answer Area	
Star	Each computer is connected by a single cable.	Topology
Mesh	Each workstation acts as a repeater.	Topology
Ring	Each computer is connected to every other computer.	Topology
	There is a central connectivity device.	Topology
	The number of connections equals the total number of computers minus one.	Topology
	Each node is connected to exactly two other nodes.	Topology

Correct Answer:

Star – Each computer is connected by a single cable.

Ring – each workstation acts as a repeater.

Mesh – Each computer is connected to every other computer

Star - There is a central connectivity device.

Mesh – The number of connections equals the total number of computers minus one.

Ring – Each node is connected to exactly two other nodes.

28

What are three advantages of VLANs? (Choose three.)

- A. They can logically address packets by using IP.
- B. They require a router to connect to VLANs on another switch.
- C. They compartmentalize a network and isolate traffic.
- D. They are efficient because a single switch can implement only a single VLAN.
- E. They act as though they are on the same LAN regardless of physical location.

Correct Answer: BCE

Correct Answer: CDE???

29

This question requires that you evaluate the underlined text to determine if it is correct.

To set lower security settings in Internet Explorer for an extranet site, add the site's URL to the "Local Intranet" zone. Select the correct answer if the underlined text does not make the statement correct. Select "No change is needed" if the underlined text makes the statement correct.

- A. Internet
- B. Trusted Sites
- C. Extranet Sites
- D. No change is needed

Correct Answer: B

30

This question requires that you evaluate the underlined text to determine if it is correct.

IPSec policies for two machines on a LAN can be modified by using the "IPSec policy snap-in" on Windows 7.

Select the correct answer if the underlined text does not make the statement correct. Select "No change is needed" if the underlined text makes the statement correct.

- A. Windows Firewall with Advanced Security snap-in
- B. LAN adapter properties
- C. Remote Access snap-in
- D. No change is needed

Correct Answer: A

31

You need to divide a network into three subnets.

Which device should you use?

- A. Hub
- B. Bridge
- C. Router
- D. Segmenter

Correct Answer: C

32

You work at a coffee shop. Your supervisor asks you to help set up a computer network.

The network needs to have the following items:

- A public facing web server
- A Wi-Fi network for customers
- A private network for the point of sale terminals
- An office PC
- A file/print server
- A network printer

You need to set up a perimeter network to protect the network.

Which two items should you include in the perimeter network? (Choose two.)

- A. Network printer
- B. Web server
- C. File server
- D. Wi-Fi network
- E. Point of sale terminals

Correct Answer: AB

33

A node within a local area network (LAN) must have which two of the following? (Choose two.)

- A. Username and password
- B. Share name
- C. NIC
- D. IP address
- E. Table of all network nodes

Correct Answer: CD

34

What are two advantages of using star topology rather than ring topology in a computer lab workgroup? (Choose two.)

- A. Failure of a central connectivity device does not bring down the entire network.
- B. A central connection point allows for flexibility and scalability.
- C. Data travels on redundant paths, so one cable cannot stop its transmission.
- D. A cable problem within the group affects two nodes, at most.

Correct Answer: BD

35

What are three characteristics of Ethernet network topology? (Choose three.)

- A. It uses tokens to avoid collisions on the network.
- B. It can use coaxial, twisted pair, and fiber optic media.
- C. It comprises the largest share of the networks in place today.
- D. It is a non-switching protocol.
- E. It can negotiate different transmission speeds.

Correct Answer: ABE

Correct Answer: BCE????

36

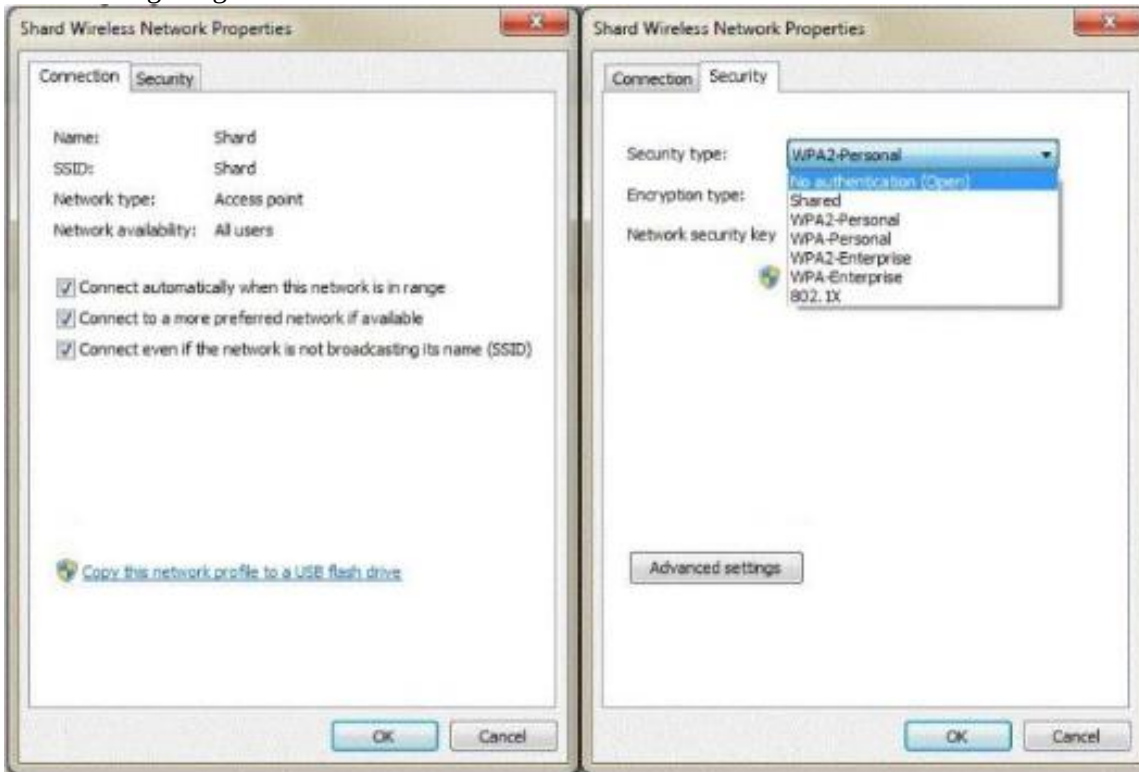
What are two characteristics of the CSMA/CD access method? (Choose two.)

- A. It checks to see if a collision has been detected.
- B. It does a round robin search for requests to transmit from all nodes on the network.
- C. It signals its intent to transmit on the network.
- D. It waits until the transmission medium is idle.

Correct Answer: AD

37

You are configuring a wireless network with the Wireless Network Properties that are shown in the following image:



Use the drop-down menus to select the answer choice that completes each statement. Each correct selection is worth one point.

Correct Answer:

To manually select which network to connect to, you should uncheck **connect to a more preferred network if available**. The **802.1X** security type requires certificates for its encryption.

38

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Each correct selection is worth one point.

	Yes	No
You use a perimeter network to grant internal clients access to external resources.	☐	☐
A LAN has no access to the perimeter network.	☐	☐
A perimeter network typically contains servers that require Internet access, such as web or email servers.	☐	☐

Correct Answer:

You use a perimeter network to grant internal clients access to external resources. - **False**

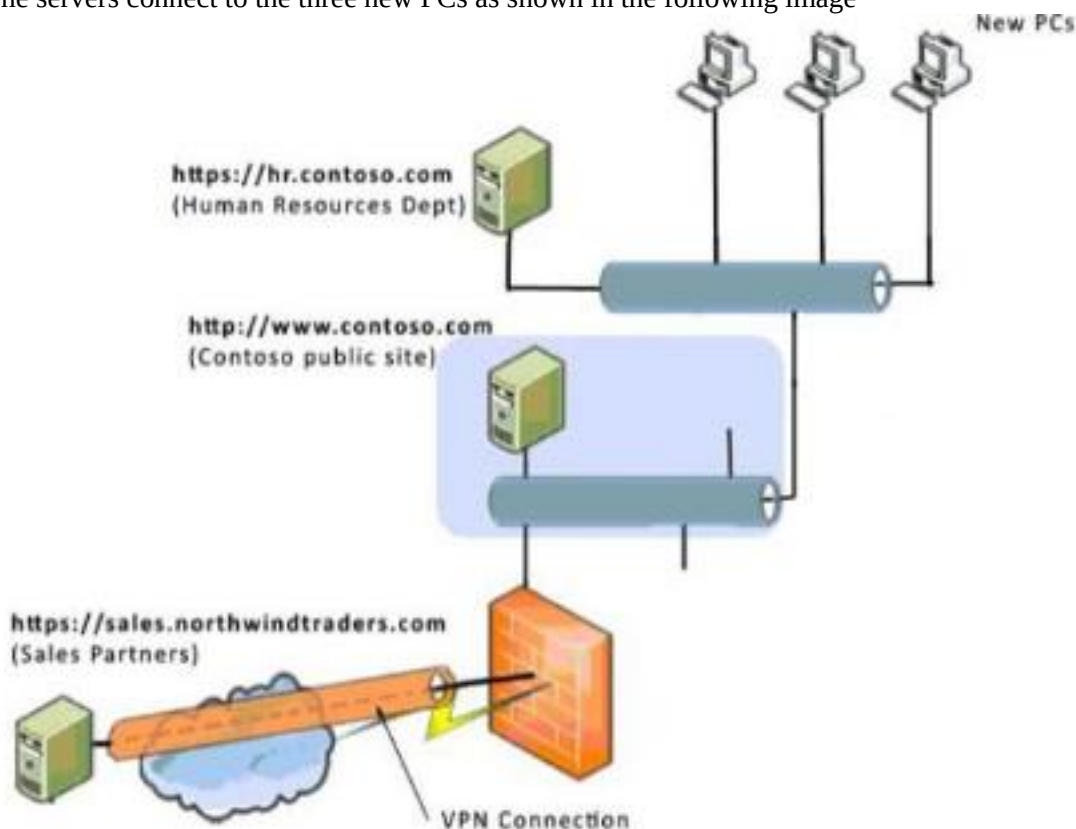
A LAN has no access to the perimeter network. - **False**

A perimeter network typically contains servers that require Internet access, such as web or email servers. - **True**

39

You are an intern for Contoso Ltd. Your supervisor asks you to configure the security zones for three new PCs so that they are able to connect to two web servers.

The servers connect to the three new PCs as shown in the following image



Use the drop-down menus to select the answer choice that completes each statement. Each correct selection is worth one point.

Correct Answer:

The appropriate type of security zone for <https://sales.northwindtraders.com> is **Trusted Sites**.

The appropriate type of security zone for https://hr.contoso.com is **Local Intranet / Trusted Sites**???

40

what device acts as a high-speed modem for LAN?

- a. analog modem
- b. telco incoming router
- c. packet switcher
- d. CSU/DSU

Correct Answer: D

2.2 Define the characteristics of wide area networks (WANs)

41

Which type of network is most vulnerable to intrusion?

- A. Dial-up
- B. Wireless
- C. Broadband
- D. Leased line

Correct: B

42

Which type of network covers the broadest area?

- A. WAN
- B. CAN
- C. LAN
- D. PAN

Correct: A

43

Which of the following is a Layer 2 WAN protocol?

- A. Point-to-Point Protocol (PPP)
- B. Simple Network Management Protocol (SNMP)
- C. Transmission Control Protocol (TCP)
- D. Internet Protocol (IP)

Correct: A

44

Which two of the following are connectivity options for wide area networks (WANs)? (Choose two.)

- A. Token ring
- B. Ethernet
- C. Dial-up
- D. Leased line

Correct Answer: CD

45

What is the bit rate for a North American T3 circuit?

- A. 6.312 Mbit/s
- B. 44.736 Mbit/s
- C. 274.176 Mbit/s
- D. 400.352 Mbit/s

Correct Answer: B

46

A university has network links between various locations. Where would a T3 connection be appropriate?

- A. Server to network in the main campus server room
- B. Main campus to a large satellite campus
- C. Computer lab PC to lab printer
- D. Library laptop PC to Internet

Correct Answer: B

Correct Answer: A???

47

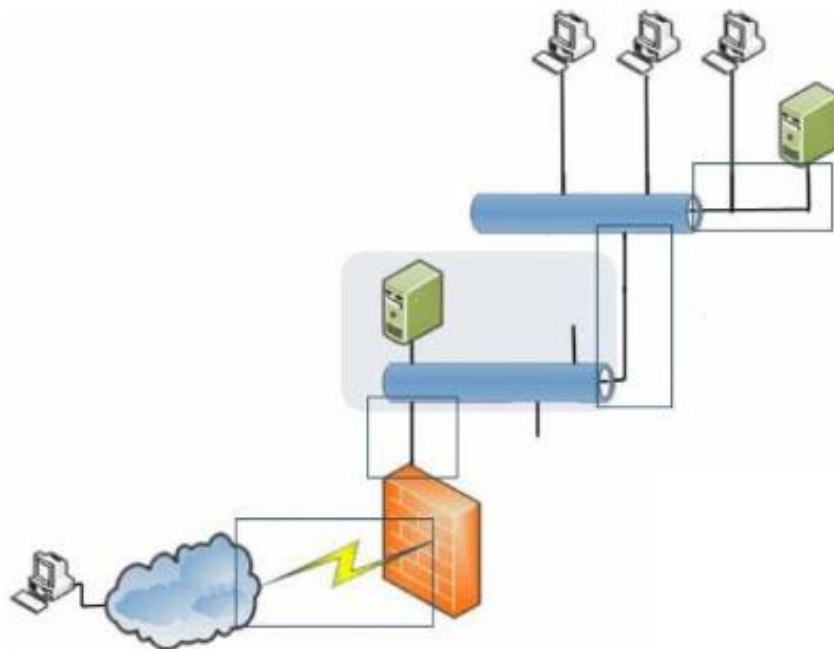
Which two features of a Windows Server 2008 R2 server should you install to use that server as a software router? (Choose two.)

- A. Network Policy and Access Services
- B. Routing and Remote Access Services
- C. Remote Administration
- D. DirectAccess

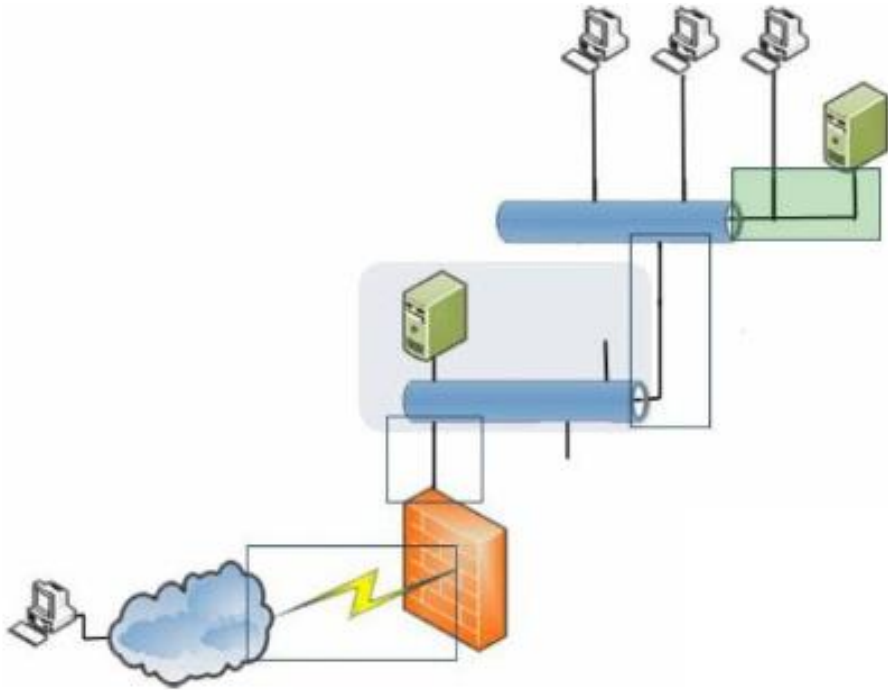
Correct Answer: AB

48

Where in the diagram is a T3 connection possible? (To answer, select the appropriate connection in the diagram in the answer area).



Correct Answer:



49

What is the primary disadvantage of X.25?

- a. digital circuit
- b. small data payloads
- c. circuit switching
- d. slow speeds

Correct Answer: A

50

The _____ is the largest WAN in the world

- a. internet
- b. intranet
- c. WWW
- d. MSDN

Correct Answer: A

51

What stage is the world wide web in?

- a. 1.0
- b. 1.1
- c. 2.0
- d. 3.0

Correct Answer: C

52

When you have multiple circuits connected to a site, the entire circuit is known as a

- a. virtual circuit
- b. emulated circuit
- c. joined circuit
- d. multitasked circuit

Correct Answer: A

53

What replaced X.25?

- a. frame relay
- b. ATM
- c. ISDN BRI
- d. DSL

Correct Answer: A

54

At what speed does a T3 line run?

- a. 1.5 Mbps
- b. 4.5 Mbps
- c. 44.7 Mbps
- d. 128 Mbps

Correct Answer: C

55

In Europe, what circuit would be similar to the T1 found in the United States?

- a. E1
- b. J1
- c. T2
- d. F1

Correct Answer: A

56

How many B channels does an ISDN PRI support?

- a. 2
- b. 8
- c. 23
- d. 48

Correct Answer: C

57

What technology uses wires from the telephone company to provide broadband internet connection?

- a. cable
- b. DSL
- c. FDDI
- d. frame relay

Correct Answer: B

58

a _____ is the guaranteed certain amount of information provided by a circuit or line.

- a. committed information route
- b. pivotal information route
- c. next hop policy route
- d. frame x frame route

Correct Answer: A

59

What speed does a T1 run at?

- a. 1.544 Mbps
- b. 2.889 Mbps
- c. 3.101 Gbps
- d. 2.54 Mbps

Correct Answer: A

60

How fast is a DS0 circuit?

- a. 32 Kbps
- b. 62 Kbps
- c. 128 Kbps
- d. 1024 Kbps

Correct Answer: B

61

X.25 and frame relay are examples of what type of WAN technology?

- a. circuit switching
- b. connection switching
- c. packet switching
- d. network switching

Correct Answer: C

62

What part of the network is the point where the responsibility of the administrator ends and the telecommunications provider's responsibility begins?

- a. last mile
- b. demarc
- c. router array
- d. PAD interface

Correct Answer: D

63

What is primary disadvantage of X.25?

- a. digital circuit
- b. small data payloads
- c. circuit switching
- d. slow speed

Correct Answer: D

2.3 Identify wireless networking methods and characteristics

64

If an 802.11g Internet connection has connectivity problems, what may be the cause?

- A. A cordless phone
- B. A cellular phone
- C. Incandescent lights
- D. Electromagnetic interference (EMI)

Answer : D

65

Which of these factors has the biggest impact on data transmission speed in a wireless network?

- A. The access method used for the network
- B. The transmission standard of the equipment used
- C. The use of strong encryption for transmissions
- D. The transmission wattage rating used on the NIC

Answer : B

66

Which of the following uses a tunneling protocol?

- A. Internet
- B. VPN
- C. Extranet
- D. VLAN

Answer : B

67

Which two are published IEEE 802.11 wireless transmission standards? (Choose two.)

- A. 802.11f
- B. 802.11g
- C. 802.11k
- D. 802.11m
- E. 802.11n

Correct: BE

68

What happens when an 802.11b node starts broadcasting within the range of an 802.11g access point?

- A. The access point will transmit, but the node will be unable to receive.
- B. A connection will be established.
- C. Both the node and the access point will be unable to transmit.
- D. The node will transmit, but the access point will be unable to receive.

Correct: B

Correct Answer: C?????

Correct Answer: D?????

69

Which technology provides the highest bit rate?

- A. T1
- B. E1
- C. DS3
- D. ISDN

Correct Answer: C

70

Which wireless authentication method provides the highest level of security?

- A. Wired Equivalency Privacy (WEP)
- B. IEEE 802.11n
- C. WI-FI Protected Access (WPA)
- D. IEEE 802.11a

Correct Answer: C

71

The maximum throughput of an 802.11g network is:

- A. 2.4 GHz.
- B. 54 GHz.
- C. 2.4 Mbps
- D. 54 Mbps.

Correct: D

72

Match each set of characteristics to the corresponding 802.11 standard.

To answer, drag the appropriate set of characteristics from the column on the left to its 802.11 standard on the right. Each set may be used once, more than once, or not at all. Each correct match is worth one point.

Sets	Answer Area
Frequency range: 2.4-2.485 Ghz Data rate: 11 Mbps	802.11a Set
Frequency range: 2.4-2.485 Ghz Data rate: 54 Mbps	802.11b Set
Frequency range: 2.4-2.485 Ghz or 5.1-5.8 Ghz Data rate: 65-600 Mbps	802.11g Set
Frequency range: 5.1-5.8 Ghz Data rate: 54 Mbps	802.11n Set

Correct Answer:

- 802.11a – Frequency range: 5.1-5.8 Ghz
Data rate: 54 Mbps
- 802.11b – Frequency range: 2.4-2.485 Ghz
Data rate: 11 Mbps
- 802.11g – Frequency range: 2.4-2.485 Ghz
Data rate: 54 Mbps
- 802.11n – Frequency range: 2.4-5.8 Ghz
Data rate: 65-600 Mbps

73

Security is a concern on wireless networks due to:

- A. The radio broadcast access method.
- B. Spread spectrum issues.
- C. Frequency modulation issues.
- D. The potential for cross-talk.

Correct Answer: A

74

This question requires that you evaluate the underlined text to determine if it is correct.

The four IEEE standards, 802.11a, b, g, and n, are collectively known as "mobile ad hoc" networks.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. WiMAX
- B. Bluetooth
- C. WiFi
- D. No change is needed

Correct Answer: C

75

Which IEEE 802.11 Wi-Fi networking standard requires dual antennas so it can run in all of its supported frequencies?

- A. 802.11a
- B. 802.11b
- C. 802.11g
- D. 802.11n

Correct Answer: D

76

This question requires that you evaluate the underlined text to determine if it is correct.

"WEP" wireless encryption is the most susceptible to interception and decryption.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct

- A. WPA-AES
- B. WPA2
- C. WPA-PSK
- D. No change is needed

Correct Answer: D

77

Which wireless communication problem is caused by electromagnetic waves?

- A. Fading
- B. Attenuation
- C. Interference
- D. Diffraction

Correct Answer: C

78

This question requires that you evaluate the underlined text to determine if it is correct.

The four IEEE standards, 802.11a, b, g, and n, are collectively known as mobile ad hoc networks.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. WiMAX
- B. Bluetooth
- C. WiFi
- D. No change is needed

Correct Answer: C

79

This question requires that you evaluate the underlined text to determine if it is correct.

In a wireless network that requires an SSL certificate, WEP handles the SSL certificate. Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. 802.1X
- B. WPA2-PSK
- C. WPA-PSK
- D. No change is needed

Correct Answer: A

80

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Each correct selection is worth one point.

	Yes	No
A wireless bridge connects Ethernet-based devices to the network.	☐	☐
A wireless bridge increases the wireless signal strength of the access point.	☐	☐
Wireless bridges always work in pairs.	☐	☐

Correct Answer:

A wireless bridge connects Ethernet-based devices to the network. – **True / False**

A wireless bridge increases the wireless signal strength of the access point - **True / False**

Wireless bridges always work in pairs. - **True / False**

81

What of the following is not a characteristic of 802.11n?

- a. frame aggregation
- b. channel bonding
- c. RFI protection
- d. MIMO

Correct Answer: C

2.4 Compare and contrast network topologies and access methods

82

Which of the following determines the media access method that is used in a network?

- A. Number of hosts connected to the network
- B. Number of domain servers on the segment
- C. Maximum speed of the media
- D. Topology and protocols

Correct Answer: D

83

In which physical network topology is each computer connected to a central point?

- A. Star
- B. Mesh
- C. Ring
- D. Bus

Correct Answer: A

84

A characteristic of the mesh topology is that it:

- A. Uses a central hub.
- B. Cannot use wired connections.
- C. Uses redundant paths.
- D. Cannot use wireless connections.

Correct Answer: C

85

What are two characteristics of a mesh network topology? (Choose two.)

- A. It is fault tolerant because of redundant connections.
- B. Every node connects to every other node on the network.
- C. It works best for networks with a large number of nodes.
- D. It requires less cabling than either a star or ring topology.

Correct Answer: AB

Correct Answer: AC????

86

What speed does FDDI use?

- a. 10 Mbps
- b. 44 Mbps
- c. 54 Mbps
- d. 100 Mbps

Correct Answer: D

87

What topology does FDDI use?

- a. bus
- b. star
- c. ring
- d. mesh

Correct Answer: C

3. Network Hardware

3.1 Describe characteristics of switches

88

A network device that associates a Media Access Control (MAC) address with a port is a:

- A. DSL modem
- B. Hub
- C. Router
- D. Switch

Correct: D

89

A Layer 2 device that connects multiple computers within a network is a:

- A. Repeater
- B. Switch
- C. Router
- D. Packet

Correct: B

90

In addition to switching, multilayer switches also:

- A. Provide Layer 3 routing functions.
- B. Interface with CAT3, CAT5, CAT5e, and fiber optics.
- C. Support 10 MB, 100 MB, and 1 GB local area network (LAN) ports.
- D. Operate by using only Layer 1 and 2 protocols.

Correct Answer: A

91

One reason to replace an unmanaged switch with a managed switch is to:

- A. Manage the routing tables.
- B. Support multiple VLANs.
- C. Reduce collision domains.
- D. Route between networks,

Correct Answer: B

92

What is an example of a network device that associates a network address with a port?

- A. Switch
- B. Router
- C. Hub
- D. DSL modem

Correct Answer: B

93

What is a similarity between Layer 2 and Layer 3 switches?

- A. Both provide a high level of security to the network.
- B. Both use logical addressing to forward transmissions.
- C. Both forward packets onto the network.
- D. Both allow the implementation of VLANs.

Correct Answer: D

Correct Answer: C????

94

Which network device interconnects computers in a workgroup, is able to be remotely configured, and provides the best throughput?

- A. Unmanaged switch
- B. Hub
- C. Router
- D. Managed switch

Correct Answer: A

Correct Answer: D????

95

What are two differences between switches and hubs? (Choose two.)

- A. Switches are slower than hubs because of the extra addressing functions that switches perform.
- B. Switches send data to all of the computers that are connected to them for efficiency.
- C. Switches are capable of sending and receiving data at the same time.
- D. Switches identify the intended destination of the data that they receive.

Correct Answer: BC

Correct Answer: AD????

96

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

	Yes	No
A switch sends unicast packets to one destination port only.	☐	☐
A switch floods ports if it does not know where to send a packet.	☐	☐
A switch sends broadcast packets to the uplink port only.	☐	☐

Correct Answer:

A switch sends unicast packets to one destination port only. - **True**

A switch floods ports if it does not know where to send a packet - **True**

A switch sends broadcast packets to the uplink port only. - **False**

3.2 Describe characteristics of routers

197

One advantage of dynamic routing is that it:

- A. Automatically maintains routing tables.
- B. Limits traffic derived from routing protocols.
- C. Reduces broadcast traffic.
- D. Automatically enables DHCP.

Correct: A

98

A router's static route is set by the:

- A. Adjacent network
- B. Next upstream router
- C. Network administrator
- D. Routing protocol

Correct: C

99

The query protocol used to locate resources on a network is:

- A. User Datagram Protocol (UDP).
- B. Lightweight Directory Access Protocol (LDAP)
- C. Tracert
- D. Telnet.

Correct: B

100

The function of a router is to:

- A. Provide IP subnet masks for hosts.
- B. Forward traffic to other networks.
- C. Broadcast routing tables to clients.
- D. Store tables for name resolution.

Correct Answer: B

101

If a router cannot determine the next hop for a packet, the router will:

- A. Forward the packet to the default route.
- B. Send the packet back to the packet's source.
- C. Broadcast the packet.
- D. Store the packet in the memory buffer.

Correct Answer: A

102

If a router is installed so that it separates a DHCP server from its clients, the clients will:

- A. Immediately lose connectivity to all segments.
- B. Be unable to obtain their leases from the server.
- C. Immediately lose connectivity to the local segment.
- D. Receive an immediate renewal of their lease.

Correct Answer: B

103

This question requires that you evaluate the underlined text to determine if it is correct.

Every router today is "TCP/IP" enabled, which is an industry-standard protocol that you use on the Internet and for local addressing.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. HTTP
- B. FTP
- C. SNMP
- D. No change is needed

Correct Answer: D

104

This question requires that you evaluate the underlined text to determine if it is correct.

"Dynamic routing" is fault tolerant.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. Static routing
- B. Default route
- C. Least cost routing
- D. No change is needed

Correct Answer: D

105

What is an example of a Layer 3 device that connects multiple computers and networks?

- A. Packet
- B. Repeater
- C. Switch
- D. Router

Correct Answer: D

106

Which metric does Routing Information Protocol (RIP) use to determine the least costly route?

- A. Delay
- B. Host ID
- C. Hop count
- D. Interface

Correct Answer: A

Correct Answer: C????

107

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

	Yes	No
Quality of Service (QoS) allows you to define the priority traffic on the network.	☐	☐
Quality of Service (QoS) allows you to control bandwidth.	☐	☐
Quality of Service (QoS) allows you to assign protocols dynamically.	☐	☐

Correct Answer:

Quality of Service (QoS) allows you to define the priority traffic on the network. - **True**

Quality of Service (QoS) allows you to control bandwidth. - **True**

Quality of Service (QoS) allows you to assign protocols dynamically. - **False**

108

_____ is a route that must be manually defined on each router

- a. static
- b. dynamic
- c. persistent
- d. global

Correct Answer: C

109

What feature within windows allows the server to act as a router?

- a. IPsec
- b. DHCP
- c. IP forwarding
- d. RDC

Correct Answer: C

110

_____ uses a clocking circuit to control the timing of communications between two WAN devices such as router

- a. static route
- b. dynamic route
- c. PPTP route
- d. poit to point

Correct Answer: A

111

What routing protocol is the most popular distance-vector routing algorithm used to determine the best routes within a network?

- a. RIP
- b. OSPF
- c. BGP
- d. IGMP

Correct Answer: A

112

What routing protocol is the most popular link-state protocol used within a large organization?

- a. RIP
- b. OSPF
- c. BGP
- d. IGMP

Correct Answer: B

113

When a packet traverses a network, each router is a _____

- a. jump point
- b. CSU/DSU
- c. jump switch
- d. hop

Correct Answer: D

3.3 Describe characteristics of physical media

114

A cable that meets the 1000BaseT standard has a maximum length of:

- A. 100 m
- B. 250 m
- C. 500 m
- D. 1,000 m

Correct: A

115

Attenuation in a wireless network signal is a result of:

- A. Number of wireless nodes connected.
- B. Distance from the access point.
- C. Interference from cellular phones.
- D. Encryption of the signal.

Correct: B

116

What is the maximum cable length for a single Cat5 UTP cable run?

- A. 285 feet/86.87 meters
- B. 328 feet/99.97 meters
- C. 432 feet/131.67 meters
- D. 600 feet/182.88 meters

Correct: B

117

Which of these cable types transmits data the greatest distance?

- A. Multi-mode fiber
- B. Single-mode fiber
- C. Cat5e
- D. Cat6

Correct Answer: B

118

To directly connect the Ethernet network interface cards (NICs) of two computers, you should use a:

- A. Crossover cable
- B. Straight cable
- C. Rollover cable
- D. Coaxial cable

Correct Answer: A

119

What is the minimum cabling requirement for a 100BaseTX network?

- A. Category 3 UTP cable
- B. Category 5 UTP cable
- C. Category 6 UTP cable
- D. Multimode fiber cable

Correct Answer: B

120

Which feature of Category 5e STP cable reduces external interference?

- A. Crosstalk
- B. Shielding
- C. Length
- D. Twisting

Correct Answer: D

Correct Answer: B????

121

You are home for vacation, and a family member wants your help to purchase and install network cable for a home office. The cable needs to support at least 300 Mbps. What is the least expensive option that meets this requirement?

- A. Cat3
- B. CAT5
- C. Cat5e
- D. Cat6

Correct Answer: C

122

The type of connector used on a 100BaseT Ethernet cable is:

- A. RJ-11
- B. RJ-45.
- C. TNC.
- D. BNC.

Correct Answer: B

123

What are two characteristics of fiber optic cable? (Choose two.)

- A. Conducts electricity
- B. Requires metal conduit
- C. Supports splicing
- D. Requires a polish for end connectors

Correct Answer: CD

124

You need to run four Ethernet network drops. Each drop is approximately 125 feet/46.33 meters.

An interference exists along the path of each drop.

You need to ensure that interference is reduced.

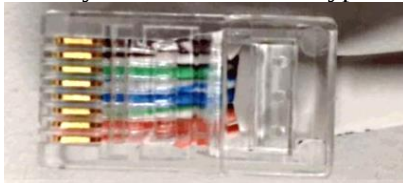
Which cable type should you use?

- A. STP Cat5e
- B. UTPCat5e
- C. Cat3
- D. UTPCat6

Correct Answer: A

125

Identify the network cable type and connector in the following graphic:



Use the drop-down menus to select the answer choice that answers each question. Each correct selection is worth one point.

Connector type	RJ45 RJ11 FDDI
Cable type	Ethernet Cat3 Fiber Optic

Correct Answer:

Connector type – RJ45

Cable type – Cat3

126

When a signal degrades as it runs through a wire, you have

- a. degradation
- b. attenuation
- c.cross over
- d.resistance

Correct Answer: D

127

What does X.25 use at its DTE device?

- a. telco incoming router
- b. packet switcher
- c. analog modem
- d. PAD

Correct Answer: D

4. Protocols and Services

4.1 Describe the Open Systems Interconnection (OSI) model

128

Which of the following represents a Media Access Control (MAC) address?

- A. GV:ZC:KK:DK:FZ:CA
- B. 255.255.255.0
- C. 05:35:AB:6E:A1:25
- D. 127.0.0.1

Correct: C

129

Order the layers of the OSI model: Select and Place:

Layers of OSI Model	Layers
Application	Layer 1
Data link	Layer 2
Network	Layer 3
Physical	Layer 4
Presentation	Layer 5
Session	Layer 6
Transport	Layer 7

Correct:

- Layer 1 - PHYSICAL
- Layer 2 - DATA LINK
- Layer 3 – NETWORK
- Layer 4 – TRANSPORT
- Layer 5 – SESSION
- Layer 6 – PRESENTATION
- Layer 7 – APPLICATION

130

In which OSI layer does routing occur?

- A. Transport
- B. Network
- C. Data Link
- D. Physical

Correct Answer: B

131

Which protocol is a transport layer protocol?

- A. FTP
- B. IP
- C. UDP
- D. ASCII

Correct Answer: C

132

Which protocol is responsible for automatically assigning IP addresses?

- A. HTTP
- B. DHCP
- C. DNS
- D. WINS

Correct Answer: A

Correct Answer: B????

133

At what layer in the OSI model are hardware addresses referenced?

- A. Network
- B. Application
- C. Data link
- D. Physical

Correct Answer: C

134

This question requires that you evaluate the underlined text to determine if it is correct.

According to the OSI model, encryption takes place on the transport layer.

Select the correct answer if the underlined text does not make the statement correct. Select "No change is needed" if the underlined text makes the statement correct.

- A. Presentation
- B. Network
- C. Application
- D. No change is needed

Correct Answer: A

135

Match each protocol to its description.
To answer, drag the appropriate protocol from the column on the left to its description on the right. Each protocol may be used once, more than once, or not at all. Each correct match is worth one point

Protocols

TCP

ICMP

ARP

UDP

IGMP

Answer Area

connectionless, message-based protocol with best-effort service

connection-oriented protocol with guaranteed service

resolves IP addresses to MAC addresses

Protocol

Protocol

Protocol

Correct Answer:
 UDP – connectionless, message-based protocol with best-effort service
 TCP – connection-oriented protocol with guaranteed service
 ARP – resolves IP address to MAC addresses

136

Match the OSI layer to its corresponding description.
To answer, drag the appropriate OSI layer from the column on the left to its description on the right. Each OSI layer may be used once, more than once, or not at all. Each correct match is worth one point.

OSI Layers

Data Link

Network

Session

Application

Answer Area

It provides network services directly to the user's application. TELNET, SMTP, and NTP operate on this layer.

It controls dialogue between source and destination nodes. RPC and NETBIOS operate on this layer.

It relies on upper layers for reliable delivery and sequencing. IPX, X.25, and NLSP operate on this layer.

It ensures that reassembled bits are in the correct order, and it requests retransmission of frames if an error occurs. Switches and WAPs operate on this layer.

It is responsible for path determination and delivery of packets, but does not guarantee delivery. ICMP, RIP, and ARP operate on this layer.

It checks for errors by adding CRC to the frame. Bridges and NICs operate on this layer.

OSI Layer

OSI Layer

OSI Layer

OSI Layer

OSI Layer

OSI Layer

Correct Answer:
 It provides network services directly to the user's application. TELNET, SMTP, and NTP operate on this layer. - **Application**
 It controls dialogue between source and destination nodes. RPC and NETBIOS operate on this layer. - **Session**
 It relies on upper layers for reliable delivery and sequencing. IPX, X.25, and NLSP operate on this layer. - **Network**
 It ensures that reassembled bits are in the correct order, and it requests retransmission of frames if an error occurs. Switches and WAPS operate on this layer. – **Data Link**
 It is responsible for path determination and delivery of packets, but does not guarantee delivery. ICMP, RIP, and ARP operate on this layer. - **Network**
 It checks for errors by adding CRC to the frame. Bridges and NICs operate on this layer. – **Data Link**

32

137

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Each correct selection is worth one point.

	Yes	No
HTTP, TELNET, FTP, and SMTP protocols operate on Layer 7 of the OSI model.	☐	☐
Layer 4 of the OSI model controls dialogue between computers.	☐	☐
Layer 3 of the OSI model controls routing between network devices.	☐	☐

Correct Answer:

HTTP, TELNET, FTP, and SMTP protocols operate on Layer 7 of the OSI model. - **True**

Layer 4 of the OSI model controls dialogue between computers. - **False**

Layer 3 of the OSI model controls routing between network devices. - **True**

138

Which layer in the OSI model covers routing between networks?

- a. physical
- b. data link
- c. network
- d. transport

Correct Answer: C

139

_____ is used to send packets from one network to another network

- a. routing
- b. transport
- c. BGP
- d. encapsulation

Correct Answer: A

4.2 Describe the Transmission Control Protocol (TCP) model

140

The protocol that maps IP addresses to a Media Access Control (MAC) address is:

- A. Internet Message Access Protocol (IMAP).
- B. Dynamic Host Configuration Protocol (DHCP).
- C. Routing Information Protocol (RIP).
- D. User Datagram Protocol (UDP).
- E. Address Resolution Protocol (ARP).

Correct Answer: E

141

Which of these is an application layer protocol?

- A. TCP
- B. FTP
- C. IP
- D. UDP

Correct Answer: B

142

An Action Center alert recommends that you enable Windows Firewall.

After enabling the firewall, you can no longer access websites.

Which two TCP ports should you add exceptions for on the firewall? (Choose two.)

- A. Port 21
- B. Port 23
- C. Port 25
- D. Port 80
- E. Port 443

Correct Answer: DE

143

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Each correct selection is worth one point

	Yes	No
The TCP/IP model has four layers, which correspond with the OSI model's seven layers.	☐	☐
The TCP/IP application layer corresponds with the top four layers of the OSI model.	☐	☐
The TCP/IP transport and Internet layers correspond with layers 3 and 4 of the OSI model.	☐	☐

Correct Answer:

The TCP/IP model has four layers, which correspond with the OSI model's seven layers. - **True**

The TCP/IP application layer corresponds with the top four layers of the OSI model. - **False**

The TCP/IP transport and Internet layers correspond with layers 3 and 4 of the OSI model. - **True**

144

Match the TCP port to the corresponding service.

Ports	Answer Area
80	SMTP Port
53	FTP Port
25	HTTPS Port
21	
443	

Correct Answer:

SMTP – 25

FTP – 21

HTTPS – 443

145

What command do you use to connect to a FTP server to download a file?

- a. nslookup
- b. ssh
- c. telnet
- d. ftp

Correct Answer: D

4.3 Describe IPv4 concepts

146

The default subnet mask for a Class B network is:

- A. 0.0.0.255
- B. 0.0.255.255
- C. 255.0.0.0
- D. 255.255.0.0

Correct: D

147

Which network does the IP address 220.100.100.100 belong to?

- A. 220.100.100.0/24
- B. 220.100.100.1/24
- C. 255.255.255.0/24
- D. 255.255.255.1/24

Correct: A

148

Which subnet mask is valid?

- A. 255.255.255.240
- B. 255.255.255.228
- C. 255.255.255.164
- D. 255.255.255.245

Correct: A

149

What is the default subnet mask for a Class C Internet network?

- A. 255.255.255.252
- B. 255.255.255.240
- C. 255.255.255.192
- D. 255.255.255.0

Correct: D

150

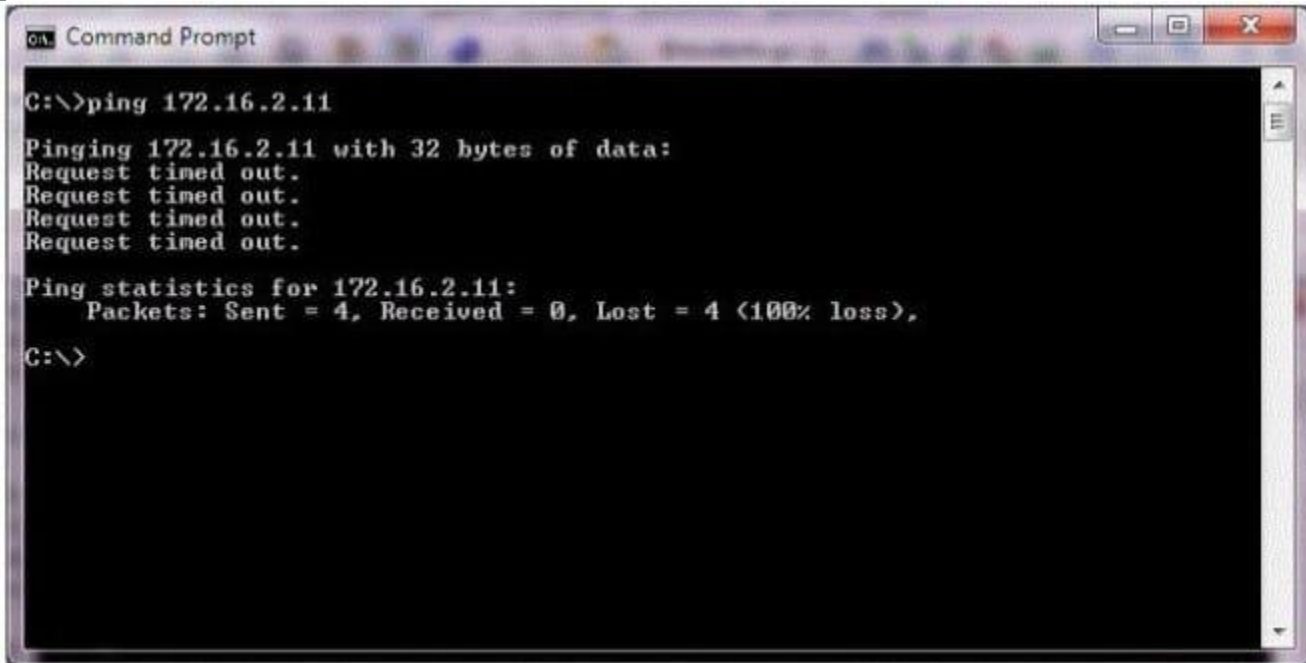
You receive a call from a family member who is unable to connect to a game server.

You learn that the server's IP is 172.16.2.11.

To help, you ping the server and receive the information shown in the following image:

Use the drop-down menus to select the answer choice that completes each statement. Each correct selection is worth one point.

Use the drop-down menus to select the answer choice that completes each statement. Each correct selection is worth one point.



```
C:\>ping 172.16.2.11

Pinging 172.16.2.11 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.16.2.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Answer Area

The server **[answer choice]** your ping requests.

▼

did not return

did return

may or may not have returned

The server status is **[answer choice]**

▼

up.

down.

unknown.

Correct Answer:

The server **did return** your ping requests.

The server **did not return** your ping requests.????

The server status is **unknown**.

151

Which of the following is a public IP address?

- A. 10.156.89.1
- B. 68.24.78.221
- C. 172.16.152.48
- D. 192.168.25.101

Correct: B

152

To which IP configuration does the CIDR notation 192.168.1.1/25 refer?

- A. 192.168.1.1 255.255.255.64
- B. 192.168.1.1 255.255.255.1
- C. 192.168.1.1 255.255.255.32
- D. 192.168.1.1 255.255.255.256
- E. 192.168.1.1 255.255.255.128

Correct Answer: E

153

The default gateway address identifies the:

- A. Device that will connect the computer to the local network.
- B. Device that will connect the computer to a remote network.
- C. Server that will provide name services for the computer.
- D. Server that will authenticate the user of the computer.

Correct Answer: B

154

This question requires that you evaluate the underlined text to determine if it is correct.

IPv4 multicast addresses range from "192.168.0.0 to 192.168.255.255".

Select the correct answer if the underlined text does not make the statement correct. Select "No change is needed" if the underlined text makes the statement correct.

- A. 127.0.0.0 to 127.255.255.255
- B. 172.16.0.0 to 172.31.255.255
- C. 224.0.0.0 to 239.255.255.255
- D. No change is needed

Correct Answer: C

155

This question requires that you evaluate the underlined text to determine if it is correct.

The loopback address of your computer is "127.0.0.1".

Select the correct answer if the underlined text does not make the statement correct. Select "No change is needed" if the underlined text makes the statement correct.

- A. 10.0.1.1
- B. 169.254.0.5
- C. 192.168.1.1
- D. No change is needed

Correct Answer: D

156

Which of these is a public address space?

- A. 192.168.0.0/16
- B. 197.16.0.0/12
- C. 10.0.0.0/8
- D. 172.16.0.0/12

Correct Answer: B

157

For each of the following statements, select Yes if the statement is true. Otherwise, select No. Each correct selection is worth one point.

	Yes	No
An IPv4 address consists of 64 bits.	☐	☐
It is standard practice to divide the binary bits of an IPv4 address into 8-bit fields named octets.	☐	☐
The value of any IPv4 octet can be from 0 to 256.	☐	☐

Correct Answer:

An IPv4 address consists of 64 bits. - **False**

It is standard practice to divide the binary bits of an IPv4 address into 8-bit fields named octets. - **True**

The value of any IPv4 octet can be from 0 to 256. - **False**

158

This question requires that you evaluate the underlined text to determine if it is correct.

IPv4 multicast addresses range from 192.168.0.0 to 192.168.255.255.

Select the correct answer if the underlined text does not make the statement correct. Select "No change is needed" if the underlined text makes the statement correct.

- A. 127.0.0.0 to 127.255.255.255
- B. 172.16.0.0 to 172.31.255.255
- C. 224.0.0.0 to 239.255.255.255
- D. No change is needed

Correct Answer: C

159

This question requires that you evaluate the underlined text to determine if it is correct.

Each IPv4 address consists of a MAC address and data-link layer address.

Select the correct answer if the underlined text does not make the statement correct. Select "No change is needed" if the underlined text makes the statement correct.

- A. Network ID and a host ID
- B. DNS record and a default route
- C. 64-bit binary number divided into octets
- D. No change is needed

Correct Answer: A

160

Match each IP address to its corresponding IPv4 address class.

IP Addresses	Answer Area
133.234.23.2	Class A IP Address
224.100.20.3	Class B IP Address
201.111.22.3	Class C IP Address
64.123.12.1	Class D IP Address

Correct answer:

- Class A – 64.123.12.1
- Class B – 133.234.23.2
- Class C – 201.111.22.3
- Class D – 224.100.20.3

161

Match the IPv4 address type to the corresponding definition.

To answer, drag the appropriate definition from the column on the left to the address type on the right. Each definition may be used once, more than once, or not at all. Each correct match is worth one point.

Definitions	Answer Area
assigned to a single network interface located on a specific subnet on the network and used for one-to-one communications	Multicast Definition
assigned to the variable portion of an IPv4 address that is used to identify a network node's interface on a subnet	Broadcast Definition
assigned to one or more network interfaces located on various subnets on the network and used for one-to-many communications	Unicast Definition
assigned to all network interface located on a subnet on the network and used for one-to-everyone-on-a-subnet communications	

Correct Answer:

Multicast – assigned to one or more network interfaces located on various subnets on the network and used for one-to-many communications
 Broadcast – assigned to all network interface located on a subnet on the network and used for one-to- everyone-on-a-subnet communications
 Unicast - assigned to a single network interface located on a specific subnet on the network and used for one-to-one communications

162

For each of the following statements, select Yes if the statement is true. Otherwise, select No. Each correct selection is worth one point.

	Yes	No
Dynamic Routing provides the ability to add networks automatically by learning them from other RIP routers.	☐	☐
Dynamic Routing provides the ability to automatically remove routes from the routing table when other RIP neighbors delete them.	☐	☐
Dynamic Routing provides the ability to select the best route based on routing metrics.	☐	☐

Correct Answer:

Dynamic Routing provides the ability to add networks automatically by learning them from other RIP routers. - **True**
 Dynamic Routing provides the ability to automatically remove routes from the routing table when other RIP neighbors delete them. - **True**
 Dynamic Routing provides the ability to select the best route based on routing metrics. - **True**

163

Match each address type to its appropriate range.

To answer, drag the appropriate address type from the column on the left to its range on the right. Each address type may be used once, more than once, or not at all. Each correct match is worth one point

Address Types		Answer Area
Loopback addresses		127.0.0.0 – 127.255.255.255 Address Type
Multicast addresses		192.168.0.0 – 192.168.255.255 Address Type
Private network addresses		224.0.0.0 – 239.255.255.255 Address Type

Correct Answer:

127.0.0.0-127.255.255.255 :Loopback addresses
 192.168.0.0-192.168.255.255 : Private network addresses
 224.0.0.0-239.255.255.255 : Multicast addresses

164

For each of the following statements, select Yes if the statement is true. Otherwise, select No. Each correct selection is worth one point.

	Yes	No
An IPv4 address consists of 64 bits.	☐	☐
It is standard practice to divide the binary bits of an IPv4 address into 8-bit fields named octets.	☐	☐
The value of any IPv4 octet can be from 0 to 256.	☐	☐

Correct Answer:

An IPv4 address consists of 64 bits. - **False**

It is standard practice to divide the binary bits of an IPv4 address into 8-bit fields named octets. - **True**

The value of any IPv4 octet can be from 0 to 256. - **False**

165

The ipconfig command will :

- Configure routers
- Configure DHCP clients
- Display a client's address
- Display a client's broadcast mode

Correct Answer: C

166

What command do you use to release the IP configuration handed out by a DHCP server?

- a. ipconfig/renew
- b. ipconfig /releasedns
- c. ipconfig /savestatic
- d. ipconfig /release

Correct Answer: D

167

What command do you use to reacquire IP configuration from a DHCP server?

- a. ipconfig/renew
- b. ipconfig /releasedns
- c. ipconfig /savestatic
- d. ipconfig /release

Correct Answer: A

168

List the four steps for a client to get an IP address from a DHCP server in the correct order

- a. discovery offering request ac knowledge
- b. request offering discovery knowledge
- c. discovery request offering knowledge
- d. request offering discovery knowledge

Correct Answer: D

169

What protocol automatically configures IP configuration for a client?

- a. DNS
- b. DHCP
- c. WINS
- d. FTP

Correct Answer: B

4.4 Describe IPv6 concepts

170

Teredo tunneling is a protocol that:

- A. Translates Internet Protocol version 4 (IPv4) to Internet Protocol version 6 (IPv6).
- B. Allows IPv6 connectivity through IPv4 devices.
- C. Provides VPN security.
- D. Dynamically allocates IPv6 addresses.

Correct: B

171

For each of the following statements, select Yes if the statement is true. Otherwise, select No. Each correct selection is worth one point.

	Yes	No
0:0:0:0:0:0:1 is the Loopback address for IPv6.	☐	☐
FECO::9C5A is a valid Site-Local IPv6 address.	☐	☐
FE80:F856:02AA is a valid Link-Local (APIPA) IPv6 address.	☐	☐

Correct Answer:

0:0:0:0:0:0:1 is the Loopback address for IPv6. - True
FECO::9C5A is a valid Site-Local IPv6 address. - True
FE80:F856:02AA is a valid Link-Local (APIPA) IPv6 address. - True

172

Which of these represents the Internet Protocol version 6 (IPv6) loopback address?

- A. 127.0.0.1
- B. 192.168.0.1
- C. FEC0:A8C0::AA01
- D. ::1

Correct Answer: D

173

Which of these addresses is a multicast address?

- A. 127.0.0.1
- B. 169.254.0.1
- C. 192.168.0.1
- D. 224.0.0.1

Correct Answer: D

174

How many bits are there in an Internet Protocol version 6 (IPv6) address?

- A. 32
- B. 64
- C. 128
- D. 256

Correct Answer: C

175

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Each correct selection is worth one point

	Yes	No
21DA:D3:0:2F3B:2AA:FF:FE28:9C5A is a valid IPv6 unicast address.	☐	☐
FE80::2AA:FF:FE28:9C5A is a valid IPv6 address.	☐	☐
21DA::02AA::FF:FE28:9C5A is a valid IPv6 address.	☐	☐

Correct Answer:

21DA:D3:0:2F3B:2AA:FF:FE28:9C5A is a valid IPv6 unicast address. - **True**
FE80::2AA:FF:FE28:9C5A is a valid IPv6 address. - **True**
21DA:02AA::FF:FE28:9C5A is a valid IPv6 address. - **False**

176

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Each correct selection is worth one point.

	Yes	No
0:0:0:0:0:0:1 is the Loopback address for IPv6.	☐	☐
FEC0:9C5A is a valid Site-Local IPv6 address.	☐	☐
FE80:F856:02AA is a valid Link-Local (APIPA) IPv6 address.	☐	☐

Correct Answer:

0:0:0:0:0:0:1 is the Loopback address for IPv6. - **True**
FEC0:9C5A is a valid Site-Local IPv6 address. - **True**
FE80:F856:02AA is a valid Link-Local (APIPA) IPv6 address. - **True**

177

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Each correct selection is worth one point.

	Yes	No
IPv6 addresses are 64-bit in length.	☐	☐
IPv6 addresses are divided into 8-bit blocks.	☐	☐
IPv6 addresses are represented by dotted-decimal notation.	☐	☐

Correct Answer:

IPv6 addresses are 64-bit in length. - **False**
IPv6 addresses are divided into 8-bit blocks. - **False**
IPv6 addresses are represented by dotted-decimal notation. - **False**

4.5 Identify well-known ports

178

The default port used for SMTP is:

- A. 23
- B. 25
- C. 80
- D. 8080

Correct: B

179

The default port used for Telnet is:

- A. 23
- B. 25
- C. 80
- D. 8080

Correct: A

180

This question requires that you evaluate the underlined text to determine if it is correct.

All session data is "encrypted between all machines" while using telnet.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. Not encrypted
- B. Encrypted between any Windows machines
- C. Encrypted only to any non-Windows machines
- D. No change is needed

Correct Answer: A

181

You are setting up a network computer game.

You need to open up ports on your firewall so your friends can join the network.

Which command displays the ports that your computer is listening for?

- A. nslookup
- B. nbtstat
- C. ping
- D. netstat

Correct Answer: A

Correct Answer: D????

182

Which of these ports does DHCP use?

- a. 80
- b. 67
- c. 23
- d. 500

Correct Answer: B

183

What port does DNS use?

- a. 443
- b. 389
- c. 51
- d. 53

Correct Answer: D

4.6 Describe name resolution concepts

184

Which setting is used to determine the Domain Name System (DNS) settings on a client computer?

- A. TELNET
- B. NSLOOKUP
- C. PATHPING
- D. NETSTAT

Correct: B

185

A service that resolves NetBIOS names to IP addresses is:

- A. Domain Name Service (DNS).
- B. Internet Service Provider (ISP).
- C. Address Resolution Protocol (ARP).
- D. Windows Internet Name Service (WINS).

Correct: D

186

What type of DNS record maps host names to addresses?

- A. Mail Exchanger (MX) DNS record
- B. Service (SRV) DNS record
- C. Host (A) DNS record
- D. Canonical (CNAME) DNS record

Correct: C

187

You ping a server by using fully qualified domain name (FQDN) and do not receive a response. You then ping the same server by using its IP address and receive a response.

Why do you receive a response on the second attempt but not on the first attempt?

- A. PING is improperly configured.
- B. The DNS is not resolving.
- C. The DHCP server is offline.
- D. NSLOOKUP is stopped.

Correct Answer: B

188

This question requires that you evaluate the underlined text to determine if it is correct.

A "secondary zone" is the first DNS zone to which all updates for the records that belong to that zone are written.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. Primary zone
- B. Stub zone
- C. Conditional forwarding zone
- D. No change is needed.

Correct: A

189

Which of the following uses pointer records and A records?

- A. IDS
- B. DNS Server
- C. NAT Server
- D. IPS

Correct Answer: B

190

The top-level domain of www.adventureworks.com is

- A. www
- B. adventureworks
- C. adventureworks.com
- D. com

Correct Answer: D

191

What type of record does DNS use to find a mail service?

- A. Service (SRV) DNS record
- B. Canonical (CNAME) DNS record
- C. Mail Exchanger (MX) DNS record
- D. Host (A) DNS record

Correct Answer: C

192

This question requires that you evaluate the underlined text to determine if it is correct.

An Address Resolution Protocol (ARP) table is used to associate IP addresses with "host names".

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. MAC addresses
- B. HomeGroup membership
- C. Preferred routers
- D. No change is needed

Correct Answer: A

193

The host name of the fully qualified domain name (FQDN) mail.exchange.corp.nwtraders.com is:

- A. corp
- B. com
- C. nwtraders
- D. exchange
- E. mail

Correct Answer: E

194

The service that resolves fully qualified domain names (FQDN) to IP addresses is:

- A. Windows Internet Name Service (WINS).
- B. Domain Name Service (DNS).
- C. Internet Service Provider (ISP).
- D. Address Resolution Protocol (ARP).

Correct Answer: B

195

This question requires that you evaluate the underlined text to determine if it is correct.

The process of replicating a zone file to multiple DNS servers is called zone replication, Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. Zone transfer
- B. Zone synchronization
- C. Start of authority
- D. No change is needed

Correct Answer: A

196

Which DNS record type specifies the host that is the authority for a given domain?

- A. NS
- B. MX
- C. CNAME
- D. SOA

Correct Answer: A

Correct Answer: D???

197

What is the DNS record type that specifies an alias name of another address record?

- A. MX
- B. CNAME
- C. NS
- D. SOA

Correct Answer: B

198

This question requires that you evaluate the underlined text to determine if it is correct.

The NTP value in a resource record indicates a length of time that other DNS servers use to determine how long to cache information for a record before expiring and discarding it.

Select the correct answer if the underlined text does not make the statement correct. Select "No change is needed" if the underlined text makes the statement correct.

- A. TTL
- B. GPS
- C. SOA RR
- D. No change is needed

Correct Answer: A

199

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

Each correct selection is worth one point.

	Yes	No
With a recursive DNS query, the DNS server will contact any other DNS servers it knows about to resolve the request.	☐	☐
When an iterative query cannot be resolved from local data, such as local zone files or a cache of previous queries, the query needs to be escalated to a root DNS server.	☐	☐
A DNS server makes an iterative query as it tries to find names outside of its local domain when it is not configured with a forwarder.	☐	☐

Correct Answer:

With a recursive DNS query, the DNS server will contact any other DNS servers it knows about to resolve the request. – **True / False???**

When an iterative query cannot be resolved from local data, such as local zone files or a cache of previous queries, the query needs to be escalated to a root DNS server. - **False**

A DNS server makes an iterative query as it tries to find names outside of its local domain when it is not configured with a forwarder. - **True**

4.7 Identify the roles of networking services

200

The ping tool is used to: (Choose two.)

- A. Determine the network portion of a host address.
- B. Self-test a host's own network interface.
- C. Determine whether a host is reachable.
- D. Manage a host's session when UDP is used.

Correct: BC

201

Which of the following are features of DHCP? (Choose two.)

- A. IP address resolution to canonical names
- B. Secure shell connections
- C. Address reservation
- D. Network file transfer
- E. IP address exclusion

Correct: CE

202

The command-line tool used to list a host's active incoming connections is:

- A. NETSTAT
- B. IPCONFIG
- C. NSLOOKUP
- D. PING

Correct: A

203

A computer that has an IP address of 169.254.0.1 cannot access the network. Which of the following services should you confirm is available?

- A. WINS
- B. DNS
- C. DHCP
- D. TFTP

Correct: C

204

When a client's DHCP-issued address expires, the client will:

- A. Select a new address and request approval from the DHCP server.
- B. Require manual configuration with a static IP address.
- C. Attempt to obtain a new address by broadcasting.
- D. Continue to use the address until it is notified to stop.

Correct: C

205

Internet Key Exchange (IKE) is responsible for which two functions? (Choose two.)

- A. Establishing network speed parameters
- B. Verifying the client's patch level
- C. Negotiating algorithms to use
- D. Exchanging key information

Correct: CD

206

Which command is used to verify that a server is connected to the network?

- A. IPCONFIG
- B. ROUTE
- C. PING
- D. CHECK

Correct Answer: C

207

The ipconfig command will:

- A. Configure routers
- B. Display a client's address
- C. Display a client's broadcast mode
- D. Configure DHCP clients

Correct Answer: B

208

To protect a network when it is connected to the Internet, you should use a:

- A. Bridge
- B. Firewall
- C. Switch
- D. Router

Correct Answer: B

209

Which of the following services masks internal IP addresses from outside the network?

- A. DHCP
- B. WINS
- C. NAT
- D. DNS

Correct Answer: C

210

Tracert is used to:

- A. Manage routing tables dynamically.
- B. Manage session-oriented connections between nodes.
- C. Report the route taken by packets across an IP network.
- D. Report the shortest route between different networks.

Correct Answer: C

211

You cannot get to any site on the Internet or on the school's intranet. Your school uses DHCP. You check your network settings, which are configured as shown in the following image:



You need to change the settings to access websites. Use the drop-down menus to select the answer choice that completes each statement.

Answer Area

Changing the IP address settings to [answer choice] will allow Internet access.

v have a default gateway address of 192.168.1.1 have a default gateway address of 192.168.13.13 obtain an IP address automatically
--

Setting the DNS server settings to [answer choice] will allow the user's computer to look up website addresses.

v obtain DNS server address automatically set Preferred DNS server to Nothing set Alternate DNS server to 192.168.13.13

Corect Answers:

Changing the IP address settings to **obtain an IP address automatically**.

Setting the DNS server settings to **obtain DNS server address automatically**.

Setting the DNS server settings to **set Alternate DNS server to 192.168.13.13.???**

212

You run the ipconfig command. The output is shown in the following image:

```
Windows IP Configuration

Wireless LAN adapter Wireless Network Connection 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wireless Network Connection:

    Connection-specific DNS Suffix . :
    IPv6 Address. . . . . : 2602:304:aecf:a929:cce2:4be1:be2a:260
    Temporary IPv6 Address. . . . . : 2602:304:aecf:a929:f09e:893:6519:6a2
    Link-local IPv6 Address . . . . . : fe80::cce2:4be1:be2a:260%17
    IPv4 Address. . . . . : 192.168.1.65
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

From these settings, you can tell that the computer:

- A. Will have limited Internet access
- B. Will have full Internet access
- C. Will not be able to access the Internet
- D. Will not be able to access the local network

Correct Answer: A

213

For which two reasons should you use IPsec between computers? (Choose two.)

- A. Data compression
- B. Data integrity
- C. Data redundancy
- D. Data confidentiality

Correct Answer: BD

214

Your home computer is having problems accessing the Internet.

You suspect that your Internet router's DHCP service is not functioning, so you check your computer's IP address. Which address indicates that your router's DHCP service is NOT functioning?

- A. 169.254.1.15
- B. 172.16.1.15
- C. 192.168.1.15
- D. 10.19.1.15

Correct Answer: A

Correct Answer: B????

215

You are helping a friend set up a public-facing web server for a home office. Your friend wants to protect the internal network from intrusion. What should you do?

- A. Set the web server in a perimeter network.
- B. Set the web server to block access on ports 80 and 443.
- C. Configure the firewall to block access on ports 80 and 443.
- D. Set the IP address of the web server to be within the LAN.

Correct Answer: A

216

A user reports that she cannot connect to network resources from a computer on the company network. The user was able to connect to the network resources yesterday. You verify that the user's computer is properly physically connected to the network. You discover that the computer's IP address is 169.254.48.97.

You need to restore access to network resources. What should you do next?

- A. Flush the cache on the DNS server.
- B. Reset the user's password on the server.
- C. Check your router's current routing tables.
- D. Verify that the DHCP service is available.

Correct Answer: D

217

Which protocol can be used to encrypt packets on the Internet?

- A. SNMP
- B. HTTPS
- C. TFTP
- D. HTTP

Correct Answer: B

218

This question requires that you evaluate the underlined text to determine if it is correct.

All session data is encrypted between all machines while using telnet.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. Not encrypted
- B. Encrypted between any Windows machines
- C. Encrypted only to any non-Windows machines
- D. No change is needed

Correct Answer: A

219

Which Microsoft network service can you use to establish a connection to a corporate LAN without any user action?

- A. VPN
- B. Remote Desktop
- C. DirectAccess
- D. Nap

Correct Answer: C

220

This question requires that you evaluate the underlined text to determine if it is correct.

When a client computer is unable to reach a DHCP server, it will automatically assign an IP address in the 10.0.0.0 - 10.0.0.255 range.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. 127.0.0.0-127.0.0.255
- B. 169.254.0.0-169.254.255.255
- C. 192.168.100.0 -192.168.100.255
- D. No change is needed

Correct Answer: B

221

This question requires that you evaluate the underlined text to determine if it is correct.

Plain old telephone service (POTS), most ISDN lines, and switched T1 lines are all examples of Message Switching.

Select the correct answer if the underlined text does not make the statement correct. Select 'No change is needed' if the underlined text makes the statement correct.

- A. Circuit Switching
- B. Packet Switching
- C. FDDI Switching
- D. No change is needed

Correct Answer: A

222

You are trying to access a music sharing service on the Internet. The service is located at the IP address 173.194.75.105. You are experiencing problems connecting.

You run a trace route to the server and receive the output shown in the following image:

```
C:\>tracert -d 173.194.75.105

Tracing route to 173.194.75.105 over a maximum of 30 hops

  0  <1 ms    <1 ms    <1 ms    10.0.0.1
  1  25 ms     29 ms     29 ms     174.57.168.1
  2   9 ms      9 ms      9 ms      68.85.76.249
  3  10 ms      9 ms      9 ms      68.86.210.25
  4  14 ms     15 ms     18 ms     68.86.92.161
  5  15 ms     16 ms     13 ms     68.86.86.142
  6  14 ms     14 ms     14 ms     75.149.231.62
  7  14 ms     15 ms     15 ms     209.85.252.80
  8  17 ms     16 ms     17 ms     72.14.236.146
  9  27 ms     28 ms     28 ms     209.85.241.222
 10  26 ms     25 ms     26 ms     216.239.48.157
 11  *         *         *         Request timed out.
 12  27 ms     26 ms     25 ms     173.194.75.105

Trace complete.

C:\>
```

Use the drop-down menus to select the answer choice that completes each statement.

Each hop in the trace route is a [answer choice]

router.

switch.

firewall.

Correct Answer:

Each hop in the trace route is a **router**.

The trace route completed **with an unknown status**.

The trace route completed **successfully**.????

The trace route completed [answer choice]

successfully.

unsuccessfully.

with an unknown status.

223

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Each correct selection is worth one point

	Yes	No
IPsec can be used to secure network communications between two machines.	☐	☐
IPsec can be used to secure network communication between two networks.	☐	☐
IPsec network traffic is always encrypted.	☐	☐

Correct Answer:

IPsec can be used to secure network communications between two machines. - True

IPsec can be used to secure network communication between two networks. - True

IPsec network traffic is always encrypted. - False

224

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Each correct selection is worth one point.

	Yes	No
The tracert command displays router addresses that are traversed between a source and a destination.	☐	☐
The tracert command determines packet loss between a source and a destination.	☐	☐
The tracert command can display a list of routers being used for all active connections.	☐	☐

Correct Answer:

The tracert command displays router addresses that are traversed between a source and a destination. - **True**

The tracert command determines packet loss between a source and a destination. - **False**

The tracert command can display a list of routers being used for all active connections. - **False**

225

What protocol is used with L2TP to provide encryption?

- a. IPsec
- b. MPPE
- c. HTTPS
- d. MSC-CHAP

Correct Answer: A

226

After you install DHCP servers and configure and activate a scope, what is the last step in installing a DHCP server?

- a. register the scope
- b. register the server
- c. authorize the server
- d. register the clients

Correct Answer: C

You are studying for finals in the student lounge. When your laptop is connected to the wireless network, access to the Internet is slow. When you plug your laptop into a wall jack, you can no longer access the Internet at all.

You run the `ipconfig /all` command. The results are shown in the following image

```
C:\>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Win7Pro
Primary Dns Suffix . . . . . :
Mode Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : domain.local

Wireless LAN adapter Wireless Network Connection:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) WiFi Link 5300 AGN
Physical Address. . . . . : 00-21-6A-1F-AA-DA
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.11.48(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Friday, May 17, 2013 9:31:02 PM
Lease Expires . . . . . : Saturday, May 18, 2013 9:31:01 PM
Default Gateway . . . . . : 192.168.11.1
DHCP Server . . . . . : 192.168.11.1
DNS Servers . . . . . : 192.168.11.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . : domain.local
Description . . . . . : Intel(R) 82567LM Gigabit Network Connection
Physical Address. . . . . : 00-24-81-E3-D4-64
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Autoconfiguration IPv4 Address. . : 169.254.143.166(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . :
NetBIOS over Tcpip. . . . . : Enabled

Tunnel adapter isatap.domain.local:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : domain.local
Description . . . . . : Microsoft ISATAP Adapter
Physical Address. . . . . : 00-00-00-00-00-00-E0
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
```

Use the drop-down menus to select the answer choice that completes each statement. Each correct selection is worth one point.

The wireless adapter has an IP address configured [answer choice]

manually.
through DHCP.
through APIPA.

The Ethernet adapter has an IP address configured [answer choice]

manually.
through DHCP.
through APIPA.

Correct Answer:

The wireless adapter has an IP address configured **through DHCP**.

The Ethernet adapter has an IP address **configured APIPA**.

228

You have a computer with an address of 169.254.32.23 and a subnet mask of 255.255.0.0, yet you cannot connect to your local file servers.

What is most likely the problem?

- a. it cannot communicate with a DHCP server
- b. the DNS servers specified are incorrect or are down
- c. NetBIOS over TCP/IP has not been enabled
- d. The network card is not connected properly to the network

Correct Answer: A

229

What protocol is used with L2TP to encrypt data?

- a. IPSec
- b. MS-CHAP
- c. MS-CHAPv2
- d. MPPE

Correct Answer: A

230

Which component of IPSec generates the encryption and authentication keys?

- a. SA
- b. AH
- c. EP
- d. MPPE

Correct Answer: A

231

Which IPSec component provides connectionless integrity and data authentication but does not provide confidentiality?

- a. SA
- b. AH
- c. ESP
- d. MPPE

Correct Answer: B

232

Which IPSec component includes the most security, including confidentiality?

- a. SA
- b. AH
- c. ESP
- d. MPPE

Correct Answer: C

233

What service on a windows network is used to translate between NetBIOS name/computer name and IP addresses?

- a. DNS
- b. WINS
- c. DHCP
- d. LDAP

Correct Answer: B

234

What provides name resolution between domain names and IP addresses

- a. DHCP
- b. DNS
- c. ARP
- d. RPC

Correct Answer: B

235

What type of network can you set up that is another company's internal network?

- a. intranet
- b. extranet
- c. internet
- d. DMZ

Correct Answer: B

236

What command do you use to display and modify the network configuration of a local computer?

- a. netsh
- b. netstat
- c. telnet
- d. nbtstat

Correct Answer: A

237

what command is used to add static routes to a windows computer?

- a. nslookup
- b. telnet
- c. route
- d. nbtstat

Correct Answer: C

238

What command do you use to display the routing table on a local system?

- a. route display
- b. route table
- c. route local
- d. route print

Correct Answer: D

239

What command do you use to display the MAC address on system?

- a. ipconfig/all
- b. pathping -m
- c. route /showmac
- d. nbstat -r

Correct Answer: A

240

What command do you use to retrieve or update your DHCP assigned configuration?

- a. ipconfig /flushdns
- b. ipconfig /all
- c. ipconfig /release
- d. ipconfig /renew

Correct Answer: D

241

What command do you use to register the computer's name and IP address with the nearest DNS server?

- a. ipconfig /renew
- b. ipconfig /renew aall
- c. ipconfig /flushdns
- d. ipconfig /registerdns

Correct Answer: D

242

What command display Ethernet statistics?

- a. netstat -e
- b. netstat -x
- c. netstat -q
- d. netstat -t

Correct Answer: A

243

What option do you use to ping constantly until you stop it?

- a. ping -t <host>
- b. ping -q <host>
- c. ping -r <host>
- d. ping -s <host>

Correct Answer: A

244

What protocol is used with L2TP to encrypt data?

- a. IPsec
- b. MS-CHAP
- c. MS-CHAPv2
- d. MPPE

Correct Answer: A

245

What server is used to translate host names to IP addresses?

- a. DNS
- b. WINS
- c. HOSTS
- d. DHCP

Correct Answer: A

246

What command can you use to connect to a mail server so that you can test SMTP?

- a. ftp
- b. nslookup
- c. telnet
- d. nbstatat

Correct Answer: C